
Observe Before Improve: From a Quarantined Minimum to an Attributed Control-Transport Failure

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Abstract

We consolidate the campaign artifacts formerly numbered 019–022 and report the result that those preregistrations could not contain prospectively. Historical direct-SSH evidence suggested that FA3 document isolation helped, but lacked immutable RunRecords. A governed reproduction subsequently progressed from mean paired $\Delta\text{val_bpb} = -0.005416$ at $n = 3$, to -0.005299 at $n = 6$, and -0.0054258 at $n = 10$, with every pair helping. This is an adoption candidate, not an adopted default. The historical campaign minimum, 0.927183, is one seed-47 treatment observation rather than a contemporaneous control. Three independent proposal agents and a separate critic therefore selected a dual-anchor recipe-transport experiment, rated 87.0/100 on explicit novelty, provenance, validity, impact, reliability, feasibility, falsifiability, and program-coherence axes. The sole immutable four-pair/eight-H200 wave was terminally invalid: a returning foreign process occupied the seed-47 treatment GPU, the watchdog identified its PID and working directory, and the arm exited 79. The deterministic evaluator returned invalid with zero admissible complete pairs. Three complete treatment endpoints—0.927454, 0.930124, and 0.927999—are descriptive only and none beats 0.927183. No replay qualification, Q2, control rebind, adoption, or new SOTA is claimed. The central result is methodological: before a system can improve itself, it must observe and attribute its failures.

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1. Scope, Supersession, and Evidence Labels

This ICML-format artifact replaces the individual live files formerly named paper 019, 020, 021, and both paper-022 variants. Their committed bytes remain recoverable in Git history. Paper 019 was the prospective provenance-repair preregistration. Paper 020 reported only the $n = 3$ checkpoint and was later made background-only because it used two rather than three independent proposers and referred to literature records not yet present in the live registry. Paper 021 repaired the proposer count and provenance, reported evidence available through $n = 6$, and contained no result from its proposed mechanisms. Its mechanism queue became background-only for block 22. Of the two paper-022 drafts, Dual-Anchor Transport Before Control was the sole canonical execution authority; Recover the Recipe, Not the Scalar was a withdrawn, non-authorizing design study.

This paper is a retrospective synthesis and authorizes no new GPU work. Repository facts, historical observations, preregistered judgments, and post-run conclusions remain distinct:

- 0.927183 is the chart and provenance anchor, not a valid control.
- Historical direct-SSH results are quarantined priors, not RunRecords.
- Prospective $n = 3$ and $n = 6$ results are promotion signals.
- Prospective $n = 10$ document isolation is an adoption candidate.
- The control-transport wave is invalid; its complete endpoints are descriptive and cannot be subsetting into an efficacy estimate.

2. Frozen Scientific Frame

All scored arms use one physical H200, STOP_MODE=time, TIME_BUDGET=300, a

Table 1. Evidence lineage for FA3 document isolation. Negative BPB deltas help. Only governed rows are causal campaign evidence.

stage	mean Δ bpb	helping	mean Δ steps
history	-0.005222	10/10	unavailable
prospective $n = 3$	-0.005416	3/3	+44.00
prospective $n = 6$	-0.005299	6/6	+43.33
prospective $n = 10$	-0.0054258	10/10	+42.6

frozen split, tokenizer and evaluator, FA3, and max-autotune-no-cudagraphs. Controls are concurrent, same-seed, and physically disjoint. Raw treatment-minus-control endpoint BPB is authoritative. The campaign’s effective standard deviation $\sigma = 0.001198$ fixes a raw minimum-effect gate

$$2\sigma = 0.002396.$$

Steps, tokens, MFU, and token-law residuals are diagnostics unless a preregistered fidelity estimand explicitly says otherwise. Moving compilation, packing, prefetch, or queue priming outside the charged clock is forbidden.

Every idea is scored from 1 (weak) to 5 (strong) on novelty (N), primary provenance (P), causal validity (V), expected local impact (I), transfer reliability (R), feasibility/cost (F), and numeric falsifiability (X). Canonical paper 022 additionally scores program coherence (C). Prestige can raise provenance; it cannot override arithmetic, implementation, dependency, fairness, or reward-hack vetoes.

3. Prospective Provenance Repair

Mechanism. Document boundaries define a block-diagonal causal graph. The FA3 varlen path receives compact cu_seqlens, preventing tokens from attending across documents and skipping cross-document blocks (Dao et al., 2024). The governed comparison changed only explicit DOC_MASK=0 versus DOC_MASK=1 under the same varlen/both-mode implementation.

At $n = 3$, seeds 53–55 respectively produced control-to-treatment BPB $0.940045 \rightarrow 0.934950$, $0.939083 \rightarrow 0.935298$, and $0.941731 \rightarrow 0.934364$, with deltas -0.005095 , -0.003785 , and -0.007367 . All arms passed source/config/runtime hashes, physical UUID and training-PID binding, pre-clock boundary attestation, co-tenancy sampling, terminal markers, and independently recomputed boundary digests. The $n = 10$ result supports the end-to-end mechanism, but does not separate attention semantics from throughput-mediated exposure or training isolation from evaluation isolation.

4. Proposal and Critique Evolution

4.1. Paper 019: provenance before novelty

Three independent proposers supplied architecture, systems, and training-dynamics candidates. The coordinator added M because the registry required prospective document-isolation reproduction. Table 2 preserves every critic score. The authorized order was $M \rightarrow E\text{-}v1 \rightarrow F \rightarrow A \rightarrow C$; these were judgments, not outcomes.

The critic removed double buffering from E-v1 because overlap is a separate causal mechanism and can move work outside the clock. Packed tables survived only if a profile attributed at least a gate-sized fraction of charged time to removable launch overhead. Correcting BOS semantics was not accepted as evidence of a gate-sized effect. ReZero supports zero-initialized residual branches, not a numerical transfer guarantee for this learned hash memory (Bachlechner et al., 2020). Full nGPT, CCE, and FA4 remained too bundled, dependency-heavy, or locally sub-gate despite credible sources.

4.2. Paper 020: broad literature, then authority correction

Paper 020 read 27 primary 2025–2026 sources and scored three proposed mechanisms, but later became background-only. The scores remain useful historical judgments:

Conditioned initialization ranked first because it was atomic and had no runtime cost, but its transfer to zero c_proj, QK/output normalization, Muon, and short training was uncertain (Saratchandran and Lucey, 2026). Curriculum ranked second because ordering evidence was confounded by source, difficulty, packing, and worktile distributions. The gate refinement was paper-derived rather than paper-faithful and violated the later mechanism-only rule.

4.3. Paper 021: three proposers and explicit vetoes

Paper 021 repaired the proposer process and corrected paper 020’s mechanism attributions. Table 4 reports every survivor or deferral; Table 5 reports every kill.

The exact-metadata mechanism removes a documented CUDA torch.nonzero host synchronization, but the fixed-batch law gives $-0.06 \ln(1.045) = -0.002641$: only 0.000245 beyond the gate. It is plausible only after a clock-faithful profile. Conditioned Q/K uses independent semi-orthogonal per-head projections; the disjoint head-offset V is explicitly a local full-rank-768 adaptation rather than a source-faithful

Table 2. Paper-019 critic ratings. Vectors are N/P/V/I/R/F/X.

ID	candidate	critic vector	disposition
M	prospective FA3 reproduction	1/5/5/5/4/5/5	rank 1, provenance repair
E	packer metadata + double buffer	5/5/4/4/3/3/5	split; keep E-v1 only
F	packed n-gram tables	4/3/4/3/3/4/5	profile-conditional keep
A	BOS-safe n-gram prefixes	4/3/5/2/3/5/5	conditional keep
C	zero-init/tanh n-gram gate	4/4/3/2/2/4/5	exploratory keep
J	stride-sampled future-2 auxiliary	5/3/3/3/2/3/5	later paper only
D	full nGPT conversion	4/5/2/4/1/1/4	defer
G	softcap-aware CCE	4/5/4/4/3/1/5	defer
B	signed CountSketch memory	4/2/1/1/1/4/3	kill
H	FA4 document masking	2/5/3/2/2/1/4	kill in block
I	epoch- ≥ 2 n-gram freeze	4/3/2/2/1/2/4	kill formulation
K	lossless tail-carry packer	4/3/5/1/3/3/5	kill
L	mask BOS targets	3/2/2/1/2/5/5	kill

Table 3. Paper-020 core proposer-to-critic ratings (N/P/V/I/R/F/X).

candidate	proposer	critic
conditioned Q/K/V	4/5/4/4/4/5/5	4/5/4/3/3/5/5
compression curriculum	4/5/4/4/4/4/4	3/5/3/3/3/3/5
elementwise head gate	2/5/5/4/4/4/5	2/5/3/2/3/4/5

claim. Zero-c_proj and normalization require a mediator test before endpoint scoring. Paper 021 also corrected curriculum attributions: 39.5% fewer steps was compression plus linear pacing, 41.7% compression plus interleaving, and 29.8% token-count plus quadratic pacing.

5. Recovering the Recipe, Not the Scalar

The historical minimum belongs to an indivisible six-coordinate recipe: TTTL attention allocation, matrix LR .04, device batch 72, total batch 147456, warm-down .85, and continuous Muon momentum. Common settings include FA3 varlen/both-mode document isolation and 64 $\times$ n-gram tables. No component receives causal credit.

The recipe mean is 0.9280515 (SD 0.00118829). Historical paired deltas have mean -0.0009590 , SD 0.00090335, and approximate 90% CI $[-0.002022, 0.000104]$. The historical pair contrast therefore neither establishes superiority nor clears the current -0.002396 gate. Relative to the frozen operational reference 0.934930, the local point prediction was -0.0068785 ; this was a design prior, not evidence.

5.1. Canonical paper-022 ranking

Three independent reproducibility, statistics, and mechanism proposers generated eight deduplicated designs. A separate critic scored N/P/V/I/R/F/X/C with fixed weights 8/17/18/14/13/9/9/12 percent.

Table 7 preserves proposer vectors, per-axis critic changes, final vectors, and weighted totals.

5.2. Why rank 1, and why it was plausible

Rank 1 simultaneously asked whether the historical recipe transported to instrumented current source and whether it beat a concurrent current recipe. Rank 2 could test present efficacy but not historical transport. Rank 3 was more attributive in principle, but the repository could not atomically govern two executable source hashes, four cells, and adequate replication. Ranks 4 and 6 failed the same multi-source/source-launch dependency. Rank 5 decomposed a sub-gate historical schedule contrast before qualifying the whole operating point. Rank 7 calibrated harness noise instead of improving the model; rank 8 was seed-47 pseudoreplication.

The causal hypothesis was connected and deliberately local. A smaller global batch yields more optimizer updates at comparable token exposure; TTTL changes attention allocation; LR, warmdown, and Muon momentum retune the finite-budget trajectory. The preserved recipe averaged 2002.75 optimizer steps and 295.318M tokens. Its point prediction had 0.0044825 BPB margin beyond the gate. No 2025–2026 paper establishes this exact bundle or predicts 0.927183. Recursive supports realistic comparator discipline and multi-seed experimentation (Recursive, 2026); Kimi K3 motivates architecture-optimization-systems co-design, not transfer of this bundle’s effect (Moonshot AI, 2026). Conditioned initialization and gated attention support later branches, not the recipe itself (Saratchandran and Lucey, 2026; Qiu et al., 2025).

6. The Canonical Q1 Experiment

Four current-source pairs used seeds 47–50 and all eight H200s. Both arms shared the frozen source and common environment. The current

Table 4. Paper-021 survivors/deferred candidates. Vectors are proposer → critic on N/P/V/I/R/F/X.

candidate	proposer → critic	disposition
exact packer metadata	5/5/5/4/4/4/5 → 4/5/5/3/3/3/5	rank 1; requires +4.5% token/s profile
conditioned Q/K + head-offset V	4/4/3/3/2/4/5 and 4/5/4/3/3/5/5 → 4/4/3/3/2/5/5	rank 2; local full-rank V adaptation
logical QKV fusion	2/5/4/3/3/4/5 → 2/5/4/2/2/4/5	rank 3; profile-conditional
charged CPU producer	3/5/3/3/2/3/5 → 3/5/4/2/2/3/5	rank 4; profile-conditional
tensorized n-gram N=3/R=512	5/5/4/4/3/2/5 → 5/3/2/3/2/1/5	defer; wrong live provenance and likely slow
compression curriculum	3/5/2/3/2/2/5 → 3/4/2/3/2/1/5	defer; score and clock wrapper unavailable
QK/V 2 × 2 attribution	3/5/5/2/3/4/5 → 2/4/5/2/3/3/5	conditional child after $n \geq 3$

Table 5. Paper-021 killed pool. All candidates retain explicit scores even when a veto overrides their totals.

candidate	proposer → critic	principal veto
full-input headwise gate	3/5/4/2/3/5/5 → 1/4/4/1/2/5/5	nearby knob
paper-020 elementwise gate	2/5/3/2/3/4/5 → 1/4/3/1/2/4/5	non-faithful refinement
memory-aware cosine n-gram gate	4/4/3/3/2/4/5 → 3/2/2/2/1/3/5	large source adaptation
packed n-gram tables	4/3/2/1/2/2/5 → 2/4/4/1/4/3/5	sub-gate profile
FA4/Flex retry	2/5/3/1/3/1/5 → 1/5/2/1/4/1/5	duplicate/missing dependency
Cut Cross Entropy	1/5/4/1/4/2/5 → unchanged	CE share only 1.626%
epoch n-gram freeze	3/3/1/2/1/2/5 → 2/4/1/1/1/2/5	invalid epoch trigger
immediate LR bracket	1/3/2/1/2/5/5 → 1/4/3/1/3/5/5	mechanism-only veto
literal repeated-identity V	critic-added 3/5/2/2/1/5/5	global rank ≤ 128
TN-gram N=5/rank sweep	critic-added 1/5/3/2/2/1/4	premature child
nonzero c_proj workaround	critic-added 1/2/1/2/1/5/5	destroys attribution

Table 6. Historical seed-47–50 block. These values are forensic anchors, not live controls.

seed	control	recipe	Δbpb	steps
47	.928310	.927183	-.001127	2022
48	.928480	.927380	-.001100	2017
49	.931671	.929782	-.001889	1983
50	.927581	.927861	+0.000280	1989

control used SSSL/.03/128/262144/.95/momentum-continuity 0; the recovered recipe used TTTL/.04/72/147456/.85/continuity 1. The decision required the intersection of:

1. a 90% replay-equivalence interval inside $[-0.002396, +0.002396]$;
2. a one-sided 95% upper bound for treatment-minus-control at or below -0.002396 , with all 4/4 pairs helping;
3. a 90% log-token-ratio interval inside $[\log(0.98), \log(1.02)]$; and
4. every source, config, runtime, device, PID, boundary, completion, and co-tenancy invariant.

Equivalence testing placed non-equivalence in the null; nonsignificance could not be relabeled as equivalence (Schuirmann, 1987; Lakens, 2017). Exact six-decimal reproduction was explicitly rejected as a reward-hack surface.

7. Terminal Result: Invalid, Observed, Attributed

Two clean node inventories preceded the immutable schedule. Seven arms completed. Late in seed-47 recovered-recipe training, the isolation watchdog observed foreign PID 2743971, parent PID 2728906, launched from /data2/ncpl-pathA/work/vbird_autoresearch/temp-epoch23-peak. The watchdog killed only the affected training PID and emitted GPU_COTENANCY_DETECTED_PID=2743971 and EXIT_CODE=79. The arm has no terminal BPB.

The deterministic evaluator returned invalid, complete_pairs=0. The exact one-wave design cannot silently drop seed 47, retry it, remap it, or promote the remaining three pairs. Relative to historical recovered endpoints, seeds 48–50 differed by +0.000074, +0.000342, and +0.000138, but these are descriptive replay residuals. None of the complete treatment endpoints beats 0.927183. Accordingly:

no transport verdict, no Q2,
no control rebind, no SOTA

This failure is scientifically useful because it is attributed. The launch was clean at a fixed commit; the later dirty flag arose from append-only wave and RunRecord ledgers. Eight remote logs were copied into a local governed artifact bundle and matched against preregistered SHA-256 values. The failure is not “vari-

Table 7. Canonical paper-022 ranking. A veto overrides the weighted total.

rank	candidate	proposer	critic δ	critic	total	disposition
1	current-source four-seed dual-anchor transport	4/5/5/5/5/4/5/5	-1/0/-1/0/-1/-1/0/0	3/5/4/5/4/3/5/5	87.0	selected after tooling
2	fresh-seed replacement funnel	3/5/5/5/5/4/5/5	0/-1/-1/0/0/0/0/-1	3/4/4/5/5/4/5/4	85.6	conditional on rank 1
3	source $\times$ recipe 2×2	4/5/5/4/5/2/5/5	+1/0/0/-1/-2/-1/0/0	5/5/5/3/3/1/5/5	82.0	launcher/sample veto
4	archived/current source bridge	4/4/4/4/4/2/5/4	-1/+1/-1/0/-1/-1/0/0	3/5/3/4/3/1/5/4	72.0	multi-source veto
5	warmdown $\times$ momentum	2/4/5/2/3/2/5/3	0/0/0/0/0/0/0/+1	2/4/5/2/3/2/5/4	70.4	deferred attribution
6	archived-source exact-recipe A/A	2/5/3/3/3/3/5/4	0/0/+1/0/0/-2/0/-1	2/5/4/3/3/1/5/3	68.8	source-launch veto
7	current-recipe A/A, $n = 16$	4/5/5/4/5/4/5/5	-2/-1/-2/-2/-1/-2/-1/-2	2/4/3/2/4/2/4/3	61.6	calibration only
8	repeated seed-47 replicas	2/4/1/2/1/5/3/1	-1/0/0/-1/0/-1/-1/0	1/4/1/1/1/4/2/1	37.4	pseudoreplication; kill

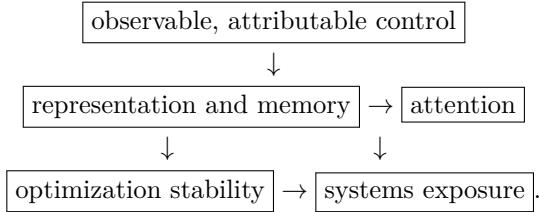
Table 8. Q1 terminal arm facts. Complete endpoints are shown for transparency; the all-four-pair estimand is invalid and no statistical gate is evaluated.

seed	arm	status	val_bpb	steps	tokens	Δ vs control	use
47	current control	complete	.933479	1177	308543488	—	descriptive
47	recovered recipe	invalid	—	—	—	—	cotenant; exit 79
48	current control	complete	.935231	1169	306446336	—	descriptive
48	recovered recipe	complete	.927454	2012	296681472	-.007777	descriptive
49	current control	complete	.937245	1144	299892736	—	descriptive
49	recovered recipe	complete	.930124	1981	292110336	-.007121	descriptive
50	current control	complete	.933191	1184	310378496	—	descriptive
50	recovered recipe	complete	.927999	1989	293289984	-.005192	descriptive

ance,” “bad luck,” or a hidden exclusion. It is a specific, timestamped resource-isolation violation whose causal consequence is a terminally missing arm.

8. Connected Improvement Program

The campaign should not respond by accumulating unrelated additions. The dependency body is:



Representation and attention share projection/value paths; attention and normalization shape residual transport; optimizer mechanisms share parameter groups; packer and producer mechanisms share loader work; systems speedups change update exposure under a wall-clock budget. Each mechanism requires observable mediators—branch norms, router entropy, gradient flow, kernel share, queue wait, or token/update exposure—so an endpoint failure can be attributed. Independent wins must precede a new prospective 2×2 interaction paper. A plateau triggers a broad, primary-literature-grounded search across representation, attention, memory, initialization, normalization, data order, optimization, and systems, followed by independent criticism; it does not license nearby knob repetition.

9. Limitations

The document-isolation effect is end-to-end and does not identify its training, evaluation, or throughput submechanisms. The recipe bundle changes six coordinates, so even a valid transport result would not attribute component effects. Four historical overlap seeds provide low-power equivalence evidence. A clean prelaunch inventory cannot guarantee that a tenant will not return later. The fixed 300-second frame rewards useful wall-clock co-design but requires unusually strict clock placement and exposure accounting. Finally, ratings are explicit judgments, not empirical effects; high scores never substitute for a successful offline gate or prospective result.

10. Conclusion

The campaign repaired one historical mechanism prospectively, reconstructed the recipe behind its best scalar, ranked control-transport designs through independent proposal and criticism, and then refused to overclaim when the selected wave lost isolation. The outcome is not a new SOTA. It is evidence that the research system can preserve a negative operational fact, identify its cause, block invalid inference, and update its program coherently. Observation and attribution are not administrative overhead: they are preconditions for self-improvement.

A. Background Literature-Pool Ratings

Table 9 preserves the paper-020 critic ratings that informed later vetoes. These are protocol/background judgments, not authorizations.

Table 9. Paper-020 background pool, part A. Vectors are N/P/V/I/R/F/X.

candidate	vector	total change	disposition	candidate	vector	total change
KernelBench-Verified	2/4/5/2/5/5/5	34→28	method	FASTKERNELS	2/4/4/2/4/4/5	33→25
SOL-ExecBench	2/4/4/2/4/4/5	32→25	method	Original KernelBench	1/5/3/1/3/3/4	28→20
FA4 H200 LPT	2/4/3/1/3/2/5	28→20	profile only	SparseTransX	1/5/3/2/3/3/5	27→22
Cut Cross Entropy	1/5/4/1/4/3/5	26→23	sub-gate	Mirage	2/5/4/1/3/1/2	26→18
Event Tensor	3/5/3/1/3/1/2	26→18	kill	ThunderKittens	2/5/4/1/3/1/3	25→19
APOLLO	2/5/3/1/3/1/3	24→18	kill	YOUMU	2/5/3/1/3/1/3	23→18
Liger	1/4/3/1/3/2/4	21→18	kill	Curriculum Dynamics	2/3/2/2/2/2/3	19→16

Table 10. Paper-020 background pool, part B.

candidate	vector	total change	disposition	candidate	vector	total change
Native Sparse Attention	4/5/4/1/4/1/4	29→23	kill	Multi-Token Attention	4/5/3/1/3/1/3	28→20
Byte Latent Transformer	4/5/5/1/4/1/3	28→23	negative	attention-side Peri-LN	2/5/3/2/3/3/5	27→23
fixed spectral conditioning	3/5/3/2/2/3/5	27→23	defer	Memory Layers at Scale	2/5/3/1/4/1/3	27→19
MuToR	3/5/4/1/4/1/4	27→22	kill	RNoPE	2/5/3/1/2/4/4	26→21
Differential Transformer	2/5/3/1/3/2/5	26→21	kill	MTP curriculum negative	1/5/5/1/5/2/5	26→24

B. Withdrawn Paper-022 Design Scores

The non-authorizing Recipe, Not the Scalar draft used N/P/V/I/R/F/X with weights .10/.15/.20/.20/.15/.10/.10. Table 11 preserves its explicit history; it must not be read as the canonical rank or execution order.

Table 11. Withdrawn paper-022 design study. Totals are out of five.

ID	candidate	proposer	critic	total change	status
A	preserved-source whole recipe	3/5/5/5/4/3/5	2/4/4/5/3/3/5	4.45→3.85	draft rank 1
B	source×recipe bridge	4/5/5/5/4/3/5	4/4/5/5/4/2/5	4.55→4.30	runner veto
C	three-arm bridge	4/5/4/5/4/4/5	3/4/3/5/3/2/5	4.45→3.65	defer
D	current-source transplant	2/3/4/5/4/5/5	2/3/3/4/3/5/5	4.05→3.50	not exact replay
E	historical schedule contrast	2/5/4/3/3/4/5	2/4/4/2/3/4/5	3.70→3.35	defer
F	old-recipe A/A	3/5/4/2/4/3/5	2/4/2/1/2/2/4	3.65→2.30	diagnostic
G	clean-directory doc-mask replay	1/5/5/5/5/4/5	unchanged	4.50→4.50	dependency
H	six-factor factorial	2/4/4/2/2/1/5	2/4/3/2/2/1/5	2.90→2.70	kill/defer
I	force 2000 steps	2/5/2/4/3/2/5	1/5/1/3/2/1/5	3.30→2.55	kill
J	shared remote source swap	1/2/1/5/1/4/3	1/1/1/4/1/4/3	2.45→2.10	kill
K	new per-arm container	3/5/5/5/5/1/5	3/4/3/4/3/1/4	4.40→3.25	defer
L	stored historical baseline	1/2/1/3/1/5/4	1/1/1/2/1/5/3	2.25→1.80	kill
M	exact scalar reproduction	1/1/1/1/1/5/1	1/2/1/1/1/5/3	1.40→1.75	kill

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